TASK 11 – Recommend and Defend the Final Design:

For this billing program, I recommend using unsigned short for the 16-bit meter reading, unsigned int for units consumed, and long long for bill amounts and the total collection. These data types are suitable for the meter readings and large bill totals in this assignment.

The tariff values should be defined in one place so that any changes can be made easily. Separate functions should be used for calculating consumption and bills, and all user inputs should be validated to prevent incorrect results. Storing money as integer paise avoids decimal rounding errors. A 64-bit long long can store the example district total of 76,000,000,000 paise.

The main limitation is that the basic design assumes at most one meter rollover between readings and fixed tariff rates. Additional rules, such as taxes or discounts, would require further changes.

If tariff values are stored in a configuration file, the program can read them when it starts. Reading t tariff values takes O(t) time and O(t) space. This allows tariff changes without recompiling the program, but the file must be checked for invalid values.